Kangjie Zhou

Contact Information

Department of Statistics
Columbia University
New York, NY 10027

https://kangjie287.github.io
kz2326@columbia.edu

Academic Positions

Columbia University — New York, NY
Founder's Postdoctoral Research Fellow — August 2024-present

Education

Stanford University — Stanford, CA
Ph.D. in Statistics, Advisor: Andrea Montanari — 2019–2024

Peking University — Beijing, China
B.S. in Mathematics, Minor in Economics — 2015–2019

Research interests

High-dimensional Statistics, Probability Theory, Deep Learning, and Causal Inference

Publications and preprints

[1] Yuchen Wu*, Kangjie Zhou*, and Weijie Su. When Does Model Collapse Occur in Structured Interactive Learning? *arXiv preprint 2605.20151,* 2026.

[2] Seunghoon Paik*, Kangjie Zhou*, Matus Telgarsky, and Ryan Tibshirani. Basic Inequalities for First-Order Optimization with Applications to Statistical Risk Analysis. *arXiv preprint 2512.24999,* 2025.

[3] Andrea Montanari* and Kangjie Zhou*. Which exceptional low-dimensional projections of a gaussian point cloud can be found in polynomial time? *arXiv preprint 2406.02970, Annals of Probability (To Appear)*, 2026.

[4] Yuhang Cai*, Kangjie Zhou*, Jingfeng Wu, Song Mei, Michael Lindsey, and Peter Bartlett. Implicit Bias of Gradient Descent for Non-Homogeneous Deep Networks. *arXiv preprint 2502.16075, ICML* 2025.

[5] Fangyi Chen*, Yunxiao Chen*, Zhiliang Ying*, and Kangjie Zhou*. Dynamic factor analysis of high-dimensional recurrent events. *Biometrika*, Volume 112, Issue 3, asaf028, 2025.

[6] Shuangping Li*, Tselil Schramm*, and Kangjie Zhou*. Discrepancy algorithms for the binary perceptron. *arXiv preprint 2408.00796, STOC* 2025.

[7] Jingyang Lyu*, Kangjie Zhou*, and Yiqiao Zhong. A statistical theory of overfitting for imbalanced classification. *arXiv preprint 2502.11323, ICLR* 2026.

[8] Raphaël Berthier*, Andrea Montanari*, and Kangjie Zhou*. Learning time-scales in two-layers neural networks. *Foundations of Computational Mathematics*, pp. 1–84, 2024.

[9] Andrea Montanari*, Yiqiao Zhong*, and Kangjie Zhou*. Tractability from overparametrization: The example of the negative perceptron. *Probability Theory and Related Fields*, vol. 188, no. 3, pp. 805–910, 2024.

[10] Yuchen Wu* and Kangjie Zhou*. Sharp analysis of power iteration for tensor pca. *Journal of Machine Learning Research*, vol. 25, no. 195, pp. 1–42, 2024.

[11] Yuchen Wu* and Kangjie Zhou*. Lower bounds for the convergence of tensor power iteration on random overcomplete models. in *Conference on Learning Theory*, pp. 3783–3820, PMLR, 2023.

[12] Andrea Montanari* and Kangjie Zhou*. Overparametrized linear dimensionality reductions: From projection pursuit to two-layer neural networks. *arXiv preprint 2206.06526*, 2022.

[13] Kangjie Zhou and Andrea Montanari. High-dimensional projection pursuit: Outer bounds and applications to interpolation in neural networks. in *Conference on Learning Theory*, pp. 5525–5527, PMLR, 2022.

[14] Kangjie Zhou and Jinzhu Jia. Propensity score adapted covariate selection for causal inference. *arXiv preprint 2109.05155*, 2021.

(*Author names ordered alphabetically)

Selected talks and presentations

- When does model collapse occur in structured interactive learning?
  ICSA China Conference, Shenzhen, China — *June 2026*
- Dynamic factor analysis of high-dimensional recurrent events
  IMS ISNPS, Thessaloniki, Greece — *June 2026*
- When does model collapse occur in structured interactive learning?
  ICSA Applied Statistics Symposium, Arlington, Virginia — *June 2026*
- Implicit bias of gradient descent for non-homogeneous deep networks
  International Conference on Statistics and Data Science, Seville, Spain — *December 2025*
- Implicit bias of gradient descent for non-homogeneous deep networks
  International Conference on Next-Generation AI and Machine Learning, San Francisco, CA — *November 2025*
- A statistical theory of overfitting for imbalanced classification
  INFORMS Annual Meeting, Atlanta, Georgia — *October 2025*
- Dynamic factor analysis of high-dimensional recurrent events
  Cornell University — *September 2025*
- A statistical theory of overfitting for imbalanced classification
  International Indian Statistical Association Conference, Lincoln, Nebraska — *June 2025*
- Dynamic factor analysis of high-dimensional recurrent events
  New England Statistics Symposium, New Haven, Connecticut — *June 2025*
- Computationally feasible projections of high-dimensional random data
  University of Wisconsin, Madison — *May 2025*
- Dynamic factor analysis of high-dimensional recurrent events
  International Conference on Statistics and Data Science, Nice, France — *December 2024*

- Dynamic factor analysis of high-dimensional recurrent events
  London School of Economics and Political Science *December 2024*
- Dynamic factor analysis of high-dimensional recurrent events
  New Jersey Institute of Technology *November 2024*
- Learning time-scales in two-layers neural networks
  Joint Statistical Meetings, Portland, Oregon *August 2024*
- Discrepancy algorithms for the binary perceptron
  Stanford CS Theory Lunch Talk *May 2024*
- Discrepancy algorithms for the binary perceptron
  MoDL Workshop at UCSD *May 2024*
- Lower bounds for the convergence of tensor power iteration on random overcomplete models
  Conference on Learning Theory *July 2023*
- Learning time-scales in two-layers neural networks
  MoDL Meeting at TTIC *May 2023*
- From projection pursuit to interpolation in two-layer neural networks
  Stanford-Berkeley Joint Colloquium *November 2022*
- High-dimensional projection pursuit: outer bounds and applications to interpolation in neural networks
  Conference on Learning Theory *July 2022*

Teaching

Teaching Assistant at Stanford University

- STATS 207 - Time Series Analysis — 2020 Spring
- STATS 202 - Data Mining and Analysis — 2020 Summer
- STATS 116 - Theory of Probability — 2020 Fall
- STATS 214/CS 229M - Machine Learning Theory — 2021 Winter
- STATS 204 - Sampling — 2021 Spring
- STATS 263/363 - Experimental Design — 2021 and 2022 Fall
- Math 230B/Stat 310B - Theory of Probability II — 2023 Winter
- STATS 218 - Introduction to Stochastic Processes II — 2022 and 2023 Spring
- STATS 141 - Biostatistics — 2024 Winter

Professional Services

- Reviewer for journals: Annals of Statistics, Journal of the American Statistical Association, Probability Theory and Related Fields, Annals of Applied Probability, Journal of Machine Learning Research, SIAM Journal on Mathematics of Data Science, Information and Inference: a Journal of the IMA, IEEE Transactions on Information Theory
- Reviewer for conferences: Conference on Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR), Conference on Learning Theory (COLT), IEEE Annual Symposium on Foundations of Computer Science (FOCS), International Conference on Algorithmic Learning Theory (ALT), International Conference on Artificial Intelligence and Statistics (AISTATS)

Industry Experience

The Voleon Group — Berkeley, CA

Research Intern — Summer 2023

Awards and Honors

- Probability Thesis Award, Stanford University 2024
- Alibaba Global Mathematics Competition, Gold medal 2021
- S.T. Yau Mathematics Competition for College Students, Gold medal in Probability and Statistics 2017
- National Scholarship, Peking University 2016–2017
- Chinese Mathematical Olympiad, First prize 2014